Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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Executive Summary

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Time-Series Forecasting

Time-series forecasting underpins operational decision-making wherever a system's future state must be anticipated from its past, from grid load to industrial condition monitoring: projecting historical sequences into future horizons is what allows a control policy to be proactive rather than reactive[1].

During the preparation of this work, the authors used AI-based tools, including GPT-5.6 Sol, Gemini 3.7 Flash, Sonnet 5, and Opus 5, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Evolution of Forecasting Methodologies

Classical time-series analysis relies on statistical models such as AutoRegressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS)[1]. These are interpretable and robust on stable low-dimensional data, but they assume linearity and stationarity and are fitted to one series at a time.

Deep learning relaxed those assumptions: recurrent networks and their gated variants[2] model non-linear dynamics through a hidden state, and training jointly across many related series established the global forecasting paradigm[3]. Transformers then replaced the recurrent bottleneck with self-attention[4], computing dependencies between distant time steps directly. Two costs follow for sampled signals: attention scales quadratically with sequence length, and a single time step is a weak unit of meaning. Patching addresses both, segmenting the series into contiguous sub-sequences and embedding each as one token, which shortens the sequence and gives every token a locally semantic support[5].

Time-series foundation models sit at the end of this line. Chronos treats time-series values as discrete tokens obtained by scaling and quantisation[6]; its successor Chronos-Bolt replaces quantisation with the patching scheme above and a direct multi-quantile head[7].[1] Patching is therefore not an incidental implementation detail, but the design choice that makes this class of models practical at scale.

Classic Aliasing

Whatever the architecture downstream, predictive fidelity is bounded by the digital representation of the signal. The governing limit is the Nyquist-Shannon sampling theorem[9]: a band-limited continuous-time signal is perfectly reconstructible only if sampled uniformly at a rate f_s strictly greater than twice its highest frequency component $f_{\max}$,

$$f_s > 2f_{\max}. \tag{1}$$

The threshold $f_N = f_s/2$ is the Nyquist frequency. When the criterion is violated, through an insufficient sampling rate or a missing anti-aliasing filter, components above f_N fold back into the lower spectrum and become indistinguishable from genuine low-frequency content[10]. This is *aliasing*, and it is irreversible: the information is destroyed at acquisition, not at inference. Respecting Equation 1 is consequently taken as sufficient for a discrete sequence to represent its source faithfully. This report asks whether that sufficiency survives patch-based tokenization.

Application Domain and Scope

The reference setting is a designed scenario: a photovoltaic array supplying an internal load, a local storage battery and the national grid, with a control agent performing load balancing, battery dispatch, state monitoring and anomaly detection on short-horizon forecasts. Measured photovoltaic and weather telemetry are taken from the SolarTech Lab dataset[11]; the concepts, states and relations of the scenario are formalised as an OWL2 ontology in Phase locking and Aliasing.

The target architecture is Chronos-Bolt-Tiny, with default patch size $P = 16$, stride $S = 16$, context window $T = 2048$, four encoder and four decoder blocks, and a direct quantile projection head[12]. The objective is to test the assumption that such a model preserves the frequency content of Nyquist-compliant signals, and to characterise how f_s, P and S interact in determining when it does not[13].

[1] The vendor reports up to a 250× inference speed-up over the original Chronos[8]; no peer-reviewed description of Chronos-Bolt has been published.

Patch-Based Tokenization: a Formal Description

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$.

The k-th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix}. \tag{2}$$

Using element-wise indexing notation,

$$\mathbf{t}_k[m] = x[kS+m], \qquad m \in \{0, \dots, P-1\}. \tag{3}$$

Before examining the degeneracies of this operator it is worth fixing what it does not do. Whenever $S \le P$, every sample of the input appears in at least one token and can be read back from it: taking $k = \lfloor n/S \rfloor$ and $m = n \bmod S \le S-1 \le P-1$ in Equation 3 gives

$$x[n] = \mathbf{t}_{\lfloor n/S \rfloor}[n \bmod S], \qquad S \le P. \tag{4}$$

Complete raw patching with $S \le P$ is therefore injective: the token sequence determines the signal, and no frequency below Nyquist is destroyed by tokenization itself. Everything that follows concerns redundancy in the token sequence rather than loss of information, the single exception being the $S > P$ regime of Case 3 below, where the operator does discard samples.

Phase locking and Aliasing

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component and model it as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{5}$$

where f denotes the signal frequency satisfying the Nyquist criterion $f < \frac{f_s}{2}$, and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S. Evaluating a stride translation yields

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \tag{6}$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of 2π-periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \qquad c \in \mathbb{N}^+. \tag{7}$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \tag{8}$$

The same derivation can be applied taking into account translational shifts equal to the patch size P, yielding a similar phase-locking condition

$$2\pi \frac{fP}{f_s} = 2\pi k \Rightarrow x[n+P] = x[n], \qquad k \in \mathbb{N}^+. \tag{9}$$

Solving Equation 7 and Equation 9 for f yields the phase-locking frequency class $\mathcal{F}_{\text{lock}}$, which is defined as the set of frequencies producing self-repetition over one stride or over one patch.

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f = c\frac{f_s}{S} \vee f = k\frac{f_s}{P},\ c, k \in \mathbb{N}^+,\ f < \frac{f_s}{2} \right\}. \tag{10}$$

It is convenient to express both conditions in units that do not depend on the sampling rate. Introducing the cycles-per-stride and cycles-per-patch ratios

$$\text{cps} = \frac{fS}{f_s}, \qquad \text{cpp} = \frac{fP}{f_s}, \tag{11}$$

Equation 7 and Equation 9 read simply $\text{cps} \in \mathbb{N}^+$ and $\text{cpp} \in \mathbb{N}^+$: $\mathcal{F}_{\text{lock}}$ is the set of frequencies completing a whole number of cycles within one stride or within one patch. This normalisation makes the two branches directly comparable across geometries and across sampling rates.

Neither branch is restricted to sinusoidal signals. Both conditions generalise to an arbitrary discrete-time signal invariant under a translation equal to the stride or to the patch,

$$x[n+S] = x[n] \qquad \text{or} \qquad x[n+P] = x[n], \qquad \forall n, \tag{12}$$

but only the first bears on consecutive patches. Tokens begin at multiples of the stride, so the step from token k to token $k+j$ is a shift of jS samples. Substituting Equation 3 and reducing by the stride condition at $j = 1$ gives

$$\mathbf{t}_{k+1}[m] = x[(k+1)S + m] = x[kS + m + S] = x[kS + m] = \mathbf{t}_k[m], \tag{13}$$

and since this holds for every component,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \tag{14}$$

The patch condition reduces the same shift only when P divides jS. At $j = 1$ that requires $P \mid S$, which under $S \leq P$ occurs only at $S = P$, so consecutive tokens are left distinct. It does hold at the smallest j for which $P \mid jS$, namely $j = P/\gcd(P, S)$, giving

$$\mathbf{t}_{k+j}[m] = x[(k+j)S + m] = x[kS + m + jS] = x[kS + m] = \mathbf{t}_k[m], \qquad j = \frac{P}{\gcd(P, S)}, \tag{15}$$

and therefore

$$\mathbf{t}_{k+j} = \mathbf{t}_k. \tag{16}$$

Patch-periodicity thus makes tokens equal as well, but at a spacing of j rather than between neighbours. The two conditions coincide on consecutive patches only in the non-overlapping geometry, where $j = 1$.

The derivations above show that some translations leave a signal unchanged; by Equation 4 this repetition is not, by itself, evidence that two distinct signals become indistinguishable to the model. Structural aliasing is therefore treated as an emergent, falsifiable property of the trained representations[13]: it is diagnosed by readability rather than by a distance between raw tokens, and it is present when the signal's frequency can no longer be recovered from the model's internal state at a locked frequency as well as it can at neighbouring, non-locked ones. $\mathcal{F}_{\text{lock}}$ is accordingly a set of candidate frequencies, derived from the geometry alone; whether the candidates are realised is an empirical question.

Analysis of Structural Redundancies and Invariances

Equation 10 is the union of two distinct families,

$$\mathcal{F}_S = \left\{ c\frac{f_s}{S} \right\}_{c\in\mathbb{N}^+}, \qquad \mathcal{F}_P = \left\{ k\frac{f_s}{P} \right\}_{k\in\mathbb{N}^+}, \qquad \mathcal{F}_{\text{lock}} = (\mathcal{F}_S \cup \mathcal{F}_P) \cap \left[0, \tfrac{f_s}{2}\right), \tag{17}$$

which coincide if and only if $S = P$, and which act on the token sequence in different ways. A frequency in $\mathcal{F}_S$ satisfies the stride condition, so by Equation 13 every token equals its predecessor: the sequence becomes one window repeated, and the number of distinct tokens drops to one however long the context. A frequency in $\mathcal{F}_P$ has a period $T_0 = f_s/f$ dividing P, so it satisfies the patch condition and by Equation 16 the sequence instead repeats every $j = P/\gcd(P, S)$ tokens, cycling through at most that many distinct vectors. Each token still holds a whole number of cycles, but neighbours read that waveform from different starting points and coincide only when $j = 1$. The two branches therefore differ in what they repeat rather than in whether they repeat: the stride branch acts between neighbouring tokens, the patch branch on the sequence as a whole.

Neither removes information. For $S \leq P$ both are regimes of Equation 4, so the signal remains recoverable and two inputs with the same token sequence are the same signal on the observed support. Both repetitions belong to the raw tokens, not to the learned representation; treating either as an invariance of the model would require showing that the patch embedding is itself shift-equivariant, which is neither established here nor assumed.

Case 1: Equal Patch Size and Stride ($P = S$) When patches are contiguous and non-overlapping the two branches coincide, $\mathcal{F}_S = \mathcal{F}_P$, and $j = P/\gcd(P, P) = 1$, so Equation 16 collapses onto Equation 14. This is the only geometry in which both conditions act on consecutive patches, and every locked frequency then gives

$$\mathbf{t}_k = \mathbf{t}_0, \qquad \forall k. \tag{18}$$

This is the most redundant geometry of the family: the only one in which every locked frequency reduces the token sequence to a single repeated vector, leaving no residual structure across tokens. It is also the default configuration of the target architecture, which operates at $P = S = 16$.

Case 2: Overlapping Windows ($S < P$) When the stride is smaller than the patch length the two branches separate. Since $f_s/S > f_s/P$, the stride comb is the sparser of the two: strictly below Nyquist it contributes $\lceil S/2 \rceil - 1$ members against the $\lceil P/2 \rceil - 1$ of the patch comb, and whenever S divides P the stride comb is a subset of the patch comb. Most locked frequencies in this regime therefore belong to $\mathcal{F}_P \setminus \mathcal{F}_S$, where consecutive tokens remain distinct and the sequence merely repeats with period $j > 1$.

Overlap does not weaken the exact condition. The derivation of Equation 13 makes no use of the relation between P and S, so at $f \in \mathcal{F}_S$ consecutive tokens are still exactly equal. What overlap changes is the behaviour around a lock: adjacent windows share $P - S$ samples by construction, so two consecutive tokens are constrained to agree on their common support regardless of frequency, and the transition into repetition is correspondingly smoother. Reducing the stride at fixed patch size therefore raises the lowest member of $\mathcal{F}_S$, thins the stride comb below Nyquist, and softens the local shape of the effect without removing it.

Case 3: Disjoint Windows ($S > P$) When the stride exceeds the patch length the windows leave $S - P$ unobserved samples between consecutive patches, and Equation 4 fails: $\mathcal{T}_{P,S}$ is non-injective for every f, since signals differing only on the gaps share a token sequence. The loss is one of discarded samples rather than of tokenization geometry, so these configurations are excluded by construction from the experimental design.

What the geometry does and does not establish The three cases can be summarised in four statements, of which only the last is an empirical claim. $\mathcal{F}_S$ collects the stride-lock candidates, at which consecutive raw tokens repeat. $\mathcal{F}_P$ collects the frequencies whose period divides the patch, at which the sequence repeats every $P/\gcd(P, S)$ tokens while neighbours stay distinct unless $S = P$. Their union $\mathcal{F}_{\text{lock}}$ is therefore a statement about candidate geometry, not a proof of aliasing: for $S \leq P$ the raw token sequence remains a lossless encoding of the input at every one of these frequencies. Structural aliasing is the separate, representation-level proposition that a model trained on such sequences nevertheless fails to make those frequencies readable from its internal state. It is a property of the learned representation rather than of the operator, and one that can only be established empirically. The hypotheses that test it are formulated in Bayesian Hypothesis.

Ontology Domain Description

To ground the experimental validation within the semantic framework established by the system ontology, we map the digital signal processing phenomena directly to the ontological layer. Let a continuous-time physical signal (such as `pv:pvPowerW` or `pv:tiltedIrradianceWm2`) be sampled uniformly at a rate f_s (`pv:samplingFrequencyHz`) and transformed into a tokenized sequence via a `pv:PatchedSignal` operator defined by patch size P (`pv:patchSize`) and translation stride S (`pv:stride`).

The architectural grid frequency f_p (`pv:patchInducedFrequencyHz`) governs the macro-temporal resolution of the token space and is defined as:

$$f_p = \frac{f_s}{S} \tag{19}$$

Based on this parametric environment, we define three distinct, falsifiable hypotheses to characterize the boundaries of structural aliasing.

Hypothesis 1 (H_1): Frequency-Dependent Blind Spots

- **Ontological Alignment:** `pv:FrequencyBlindSpot` (asserted via SWRL inference rules `R18` and `R19`).
- **Statement:** When the fundamental frequency f (`pv:fundamentalFrequencyHz`) of an input time-series signal matches an integer multiple of the patch-induced frequency f_p, the tokenization layer maps distinct cyclical variations into static, identical latent representations, creating discrete informational blind spots across the spectrum.
- **Mathematical Boundary:** An aliasing event is triggered when the frequency ratio satisfies the harmonic condition:

$$f = n \cdot f_p = n\left(\frac{f_s}{S}\right), \quad n \in \mathbb{N}^+ \tag{20}$$

  Under this boundary condition, the column rank of the local trajectory matrix collapses: $\mathrm{rank}(\mathbf{M}) = 1$ under full integer phase-locking, and $\mathrm{rank}(\mathbf{M}) \leq q$ under rational phase-locking with denominator q (see Section).
- **Falsifiability Criterion:** This hypothesis is falsified if empirical probing models or closed-loop reconstruction tasks demonstrate uniform representation fidelity across the operational band $[2, \frac{f_s}{2}]$ Hz, without exhibiting statistically significant localized performance degradation or info-loss anomalies at the exact integer harmonics of f_p.

Hypothesis 2 (H_2): Phase Alignment and Temporal Displacement

- **Ontological Alignment:** `pv:PhaseAliasingEffect` (asserted via SWRL inference rule `R20`).
- **Statement:** The magnitude of structural information loss at an aliased harmonic frequency is non-deterministic and varies as a function of the temporal alignment between the signal's peak-trough dynamics and the rigid boundaries of the tokenization grid.
- **Mathematical Boundary:** For a fixed harmonic frequency $f \in \mathcal{F}_{\text{lock}}$ and phase offset $\phi \in [0, 2\pi)$ in the signal

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{21}$$

the phase offset ϕ (radians) displaces the signal by

$$\tau = \frac{\phi\, f_s}{2\pi f} \quad \text{(samples)} \tag{22}$$

relative to the patch grid origin. Structural misalignment occurs when $\tau \bmod S \neq 0$, equivalently when

$$\phi \notin \left\{ 2\pi k \cdot \frac{f\, S}{f_s} : k \in \mathbb{Z} \right\}. \tag{23}$$

- **Falsifiability Criterion:** This hypothesis is falsified if information-theoretic tracking metrics (such as the Space Saving metric SV) or empirical probing classification accuracies remain strictly invariant across all randomly generated phase displacements $\phi \sim \text{Uniform}(0, 2\pi)$ at a locked harmonic frequency, proving that boundary location exercises no control over latent feature degradation.

Hypothesis 3 (H_3): Geometric Configuration and Overlap Mechanics

- **Ontological Alignment:** `pv:OverlapEffect` (asserted via SWRL inference rule `R21`).
- **Statement:** Altering the geometric relationship between the patch window length and the translational step size shifts the failure mode of the token space, moving the representation from absolute vector degeneracy to high-dimensional contextual redundancy.
- **Mathematical Boundary:** The geometric interaction is dictated by the token overlap ratio *or* (`pv:overlapRatio`):

$$or = \frac{P - S}{P} \tag{24}$$

  When $or = 0$ ($P = S$), the windowing is strictly contiguous and non-overlapping; satisfying Equation (2) forces consecutive tokens to be mathematically identical ($\mathbf{t}_j = \mathbf{t}_{j+1}$), wiping out temporal gradients. When $or \to 1$ ($S \ll P$), successive patches share massive historical context; although the stride-bound phase-locking condition $x[n + S] = x[n]$ still holds, the expanded patch dimension P forces tokens to capture intra-patch phase variations that break the absolute vector identity, shifting the error into an over-smoothed informational redundancy.
- **Falsifiability Criterion:** This hypothesis is falsified if varying the architectural parameters P and S while keeping a locked signal frequency f yields an identical trajectory matrix rank profile and an identical distribution of downstream transformer attention weights, proving that the geometric layout of the patch grid does not modulate the structural aliasing expression.

Bayesian Hypothesis

Each hypothesis receives its own measurement, its own model and its own estimand. Table 1 lists the models with the value each claim predicts, Table 2 every prior, and Table 3 the rule that turns a posterior into a verdict. All are ordinary regressions; only the quantity on the left and the question the coefficient answers change between them.

Each model is built the same way: a measurement, a linear predictor μ_i that adds one offset per group the observation belongs to, and a likelihood. One coefficient is the estimand and the rest keep it unconfounded. Table 2 gives every prior with the role its parameter plays. The range over which the prior scale is varied as a sensitivity check is $\{0.25, 0.5, 1.0\}$; all are centred on no effect, so a posterior that has not moved from its prior is itself a report of an absent effect.

H1

The behavioural measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$, collected in matched triplets: each lock f_k with its two controls, all three sharing the same background realisation and the same phase, which is what makes the contrast paired. The contrast used is

Model	Claim	Estimand	Predicts	Input
A	$H1$ behavioural	$e^{\bar{\beta}}$: recovery at a lock / at its controls	< 1	`contrasts`
B	$H1$ representational	$e^{\theta_{lock}}$: codelength at a lock / elsewhere	> 1	`mdl_cells`
C	$H2$	σ_ϕ: spread of the deficit across phase	≈ 0	`contrasts`
D1	$H3$ location	θ_S, θ_P: dip depth on each grid, by LOO	both < 0	`collapse`
D2	$H3a$ stride branch	κ_S: measured / stride-predicted spacing	$= 1$	`sites`
D2	$H3b$ patch branch	κ_P: measured / patch-predicted spacing	$= 1$	`sites`
A′	$M1$ mitigation	δ_O: change in the deficit per unit of overlap	< 0	`contrasts`

Table 1: *The models, the quantity whose posterior decides each claim, the value the claim predicts, and the probing table each is fitted to. $H1$ is tested twice, on the forecast (A) and on the representation (B), so that a discrepancy between them stays visible. $H3$ is split into its two branches, since $\mathcal{F}_{lock}$ is a union and a frequency may belong to either. $M1$ is not a separate model but the configuration level of A, promoted to a named estimand.*

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right], \tag{25}$$

one number per triplet, negative when the lock recovers less than its neighbours. It can be formed only because the injected tone has a known frequency. Model A fits it as

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big). \tag{26}$$

Equation 26 has four components, each motivated separately.

- **Three offsets in μ_i:** β_c per configuration, u_k per lock harmonic and u_b per background draw, so that neither an unusual frequency nor a single corpus can be mistaken for the effect.
- **Student-t_4, not Gaussian.** Where the model rebuilds nothing, recovery is near zero and the logarithm returns very large contrasts; under a Gaussian a small number of them would determine the estimate.
- **β_c drawn around $\bar{\beta}$**, making the geometries instances of one phenomenon rather than unrelated experiments. Two consequences follow. $\bar{\beta}$ exists as a parameter, which $H1$ requires, being a claim about patch-based tokenisation and not about a single checkpoint; and each geometry contributes equal weight, so the largest design cannot dominate the estimate when the number of usable triplets varies several-fold with the in-band lock count.
- **Two configuration-level covariates**, estimable together rather than confounded because the design varies overlap and patch size independently. Overlap enters as a ratio, patch size as $\log P$ because the patch grid has spacing f_s/P and equal steps in $\log P$ are equal ratios of spacing. The tilde denotes centring on the design, $\widetilde{O}_c = (O_c - \bar{O})/0.5$ and $\widetilde{\log P}_c = \log P_c - \overline{\log P}$, which leaves $\bar{\beta}$ the effect at the average configuration rather than an extrapolation to $O = 0$ and $P = 1$.

Model B poses the same question of the representation, on a frequency-local prequential codelength: at each f_c a probe separates tones at $f_c \pm 1\,\text{Hz}$ from a captured internal state. Ten states are probed: the encoder [REG] token after each encoder block, the decoder state after each decoder block, the pre-projection vector and the quantile head; the decoder-side vectors are named decoder states because Chronos-Bolt carries [REG] in the encoder only. The regression is Gamma with a log link, so $e^{\theta_{lock}}$ is the factor by which the codelength of a locked frequency increases, with probe stage and geometry as zero-mean offsets since codelengths differ several-fold between stages. The band tasks summarise hundreds of frequencies at once, so the lock indicator has nothing to attach to; they are kept as a cross-check.

Model	Parameter	Prior	Role
A, C	$\bar{\beta}$	Student-$t_4(0, 0.5)$	population phase-lock effect, on the log-recovery scale
A	δ_O	Student-$t_4(0, 0.5)$	slope in $\widetilde{O}_c$, the centred overlap; one unit $= 0.5$ of O
A	δ_P	Student-$t_4(0, 0.5)$	slope in $\widetilde{\log P}_c$, the centred log patch size
A, C	$\tau, \sigma_k, \sigma_b, \sigma$	Half-Student-$t_4(0, 0.5)$	configuration, harmonic, background and residual scales
C	σ_ϕ	Half-$\mathcal{N}(0, 0.25)$	spread of the eight per-phase offsets
B	α_0	$\mathcal{N}(\log \bar{y}, 1)$	baseline log codelength
B	θ_{lock}	$\mathcal{N}(0, 0.5)$	log codelength expansion at a lock
B	k	Gamma$(2, 0.1)$	Gamma shape (dispersion)
B	$\sigma_{st}, \sigma_{geo}$	Half-$\mathcal{N}(0, 1)$	spread of the probe-stage and geometry offsets
D1	$\alpha_g, \theta_S, \theta_P$	$\mathcal{N}(0, 1)$	per-geometry off-grid level and the two grid labels
D1	σ	Half-$\mathcal{N}(0, 1)$	residual scale of $\log z_g(f)$
D2	κ_S, κ_P	$\mathcal{N}(0, 1)$	measured spacing per unit of predicted spacing
D2	σ_F	Half-$\mathcal{N}(0, 5)$ Hz	scatter beyond the sweep-resolution floor Δf

Table 2: *Every parameter with its prior and its role. The effect priors are those of Deliverable 1, extended by δ_P, which the design of Table 4 makes estimable.*

H2

$H2$ uses the same triplets with one change. Equation 25 has already averaged over phase, so the phase index is kept instead and the response becomes the per-phase deficit $y_i = \log(R_{\text{ctrl},i} + 0.01) - \log(R_{\text{lock},i} + 0.01)$, which is d with its sign reversed and positive when the lock is worse. Model C is Model A on that response, without the configuration-level covariates and with the phase circle cut into eight slices, each given its own offset u_p:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{27}$$

The estimand σ_ϕ is the spread of those eight offsets, and it answers the question with one number. If the deficit depends on where in its cycle the signal starts, the offsets differ and σ_ϕ is large. If the geometry alone determines it, they are equal and σ_ϕ is near zero, as $H2$ predicts. Slices are used rather than a fitted curve because they assume nothing about the form a dependence might take. Dropping u_p gives a nested null, and comparing the two by leave-one-out cross-validation (LOO) separates the size of the phase effect from the evidence that it exists.

H3

$H3$ was stated as two claims, one per branch. Both are tested on the token collapse $z_g(f)$, the dispersion of the input-patch embeddings across patches, which is zero when consecutive patches are identical. All geometries share one sweep grid: a uniform 1 Hz sweep unioned with the predicted sites of every configuration swept. No geometry is therefore scored only where its own theory predicts a dip, and non-integer sites such as $42.\overline{6}$ Hz for S=12 are covered.

The first question is where the dips are located. Each swept frequency carries two labels, one per branch:

Claim	Rule	Prior	Reads as
$H1$ supported	$\Pr(\bar{\beta} < \log 0.8 \mid D) \geq 0.95$	0.34	at least 20% attenuation at a lock
$H1$ refuted	$\Pr(\lvert\bar{\beta}\rvert < \log 1.1 \mid D) \geq 0.95$	0.14	any effect is smaller than 10%
$H2$ supported	$\Pr(\sigma_\phi < \log 1.1 \mid D) \geq 0.95$	0.30	phase moves the deficit by under 10%
$H3$ location	two-label fit wins LOO, $\theta_S, \theta_P < 0$	—	dips sit on both predicted grids
$H3a$, $H3b$	$\Pr(\lvert\kappa_F - 1\rvert < 0.1 \mid D) \geq 0.95$	0.05	that branch tracks its own parameter
$M1$ supported	$\Pr(\delta_O < 0 \mid D) \geq 0.95$, overlap term wins LOO	0.50	overlap reduces the deficit

Table 3: *Decision rules.* ***Prior*** *is the probability each rule already scored under the priors of Table 2: a posterior counts as evidence only in as much as it has moved away from that value. A claim is supported at* 0.95*, refuted at* 0.05*, and otherwise inconclusive.*

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \text{OnStrideGrid} + \theta_P \cdot \text{OnPatchGrid},\ \sigma)\,. \tag{28}$$

The logarithm makes the labels multiplicative on the per-geometry off-grid level α_g. The two label coefficients θ_S and θ_P are the estimands, and $H3$ predicts both negative, so the two-label fit should win the LOO comparison against each label alone and against neither. The two labels are not competing explanations of the same dips, since a frequency may carry either or both. The two coefficients separate, however, only where exactly one label applies, which for θ_S means the geometries with $S \nmid P$.

The second question is whether the dips move. The fit above remains compatible with dips that merely sit on a fixed grid, so each branch is checked against its own scaling law. Every detected dip is assigned to the branch that predicts it, and sites belonging to both are set aside. The fundamental $\hat{f}^F_{1,g}$ of branch F in geometry g is then compared with the spacing that branch predicts, $\Delta^S_g = f_s/S_g$ or $\Delta^P_g = f_s/P_g$:

$$\hat{f}^F_{1,g} \sim \mathcal{N}\left(\kappa_F\,\Delta^F_g,\ \sqrt{\sigma_F^2 + \Delta f^2}\right). \tag{29}$$

The slope κ_F is the estimand, measured spacing per unit of predicted spacing, so $\kappa_F = 1$ denotes a branch that tracks its own arithmetic. The scatter σ_F about it is given a floor Δf, the sweep step, since a spacing read off a discrete grid is no more precise than one step; without the floor, sites landing exactly on the prediction drive σ_F to zero and the posterior degenerates.

***H*3*a*, the stride branch** The prediction is $\kappa_S = 1$: the stride sites track f_s/S one for one, so a site that remains fixed while S changes refutes it. Identification needs a series that holds P fixed and varies S, which Table 4 supplies at P=16, 24 and 32.

***H*3*b*, the patch branch** The prediction is $\kappa_P = 1$ in the same way. Identification needs a series that holds S fixed and varies P, which Table 4 supplies at S=8, 12 and 16.

Decisions and validation

Every threshold in Table 3 is fixed before a posterior is inspected, and validation comes before interpretation. Models A and C include a per-background random offset u_b with scale σ_b; this lets the posterior separate generator-specific variation from the aliasing effect itself, so that a recovery difference between Light TSMixup and KernelSynth backgrounds is attributed to the generator rather than inflating or deflating the estimate of $\bar{\beta}$. The experimental grid, sampling parameters and generators are described in Signal Generation and Testing.

Notebook workflow

The Bayesian analysis is implemented in one reproducible notebook. Its role is not to introduce a new method after the fact, but to execute the hypotheses, measurements, priors and decision rules defined above in a fixed order. The notebook therefore has a short setup stage followed by five inferential parts: priors, observation, likelihood, posterior inference and checks. Each part writes checkpoints to disk, so that an interrupted run can resume without changing the statistical design.

Part 0 (Setup). The notebook installs only missing dependencies, locates the repository, and creates the checkpoint directories. The design constants, including sampling rate, analysis band, context window, prediction horizon and the retrained geometries of Table 4, are imported from a shared probing library as described in Signal Generation and Testing.

The Bayesian sampler that will be used in Parts 3–5 is also configured here: the NUTS algorithm explores the posterior distribution with 4 independent chains, each discarding 2000 initial warm-up steps used to calibrate the step size and retaining the subsequent 2000 draws as posterior samples; `target_accept` $= 0.9$ controls the sampler's step-size adaptation, with higher values producing more cautious but more precise exploration.

Part 1 (Priors). Before any Chronos output is loaded, the notebook builds the design skeleton: one planned row per geometry, lock frequency, background realisation, generator and phase. It then declares the priors of Table 2, the decision thresholds of Table 3, and the prior-sensitivity ladder $\{0.25, 0.5, 1.0\}$, three scales at which the prior width of Model A will be varied in Part 5 to verify that the H1 conclusion is driven by the data rather than by the choice of scale. Prior predictive checks simulate data from the prior alone, without any Chronos output, to verify that the declared priors are centred on no effect, admit effects large enough to falsify or support the hypotheses, and make the decision thresholds reachable from both sides. The prior specification is saved as both JSON and Parquet, making later drift between the notebook and the report detectable.

Part 2 (Observation). A collection module loads, for each geometry, the retrained Chronos-Bolt weights, generates the synthetic signals, runs the predictions, and writes five tables in which each row is one observation and each column is one variable:

- **`contrasts`**: one row per triplet (lock frequency f_k with its two controls $f_k \pm \delta$), sharing background realisation and phase. Each row carries the forecast amplitude recovery R at the three frequencies and the contrast d. Feeds Models A and C; the phase index is retained for H2.
- **`mdl_cells`**: one row per probed frequency f_c, probe stage and geometry. Each row carries the prequential codelength of a linear probe separating $f_c - 1\,\mathrm{Hz}$ from $f_c + 1\,\mathrm{Hz}$. Feeds Model B.
- **`mdl_bandtasks`**: one row per band task, geometry and probe stage. Retained only as a descriptive cross-check of overall decodability; not used by any Bayesian model.
- **`collapse`**: one row per swept frequency and geometry. Each row carries the across-patch token dispersion $z_g(f)$. Feeds Model D1.
- **`sites`**: one row per detected collapse site and branch. Each row carries the fundamental f_1 and the measured spacing $\hat{\Delta}$. Feeds Model D2.

Collection is checkpointed per geometry: a completed geometry is reloaded rather than recomputed, so an interrupted run resumes from the last incomplete geometry without altering the statistical design. The generated TSMixup and KernelSynth backgrounds are archived alongside the results, so that the exact signals behind any posterior can be inspected.

Part 3 (Likelihood). The notebook defines the five statistical models (A, B, C, D1, D2) as PyMC specifications, each built from the corresponding table of Part 2. A shared sampling wrapper fits every model with the same NUTS settings declared in Part 0, so that no model receives a more or less careful exploration than another. Before fitting the real observations, a parameter-recovery test is run on Model A, the central and most complex model. Synthetic contrasts are generated from Model A itself on the real experimental structure (the same geometries, lock frequencies, backgrounds and phases that the actual data follow), fixing the population effect $\bar{\beta}$ at four known values: 0 (no attenuation), $\log 0.8$ (20%), $\log 0.5$ (50%) and $\log 0.1$

(90%). For each scenario the model is refitted on these synthetic data and the posterior of $\bar{\beta}$ is compared against the known value. If the posterior recovers it, the experimental design has enough statistical power to detect the effect or mark a genuine absence rather than insufficient design.

Part 4 (Posterior inference). Each model is fitted with NUTS or reloaded from disk. For each hypothesis the notebook reports the 95% credible interval of the estimand, the posterior probability of the decision threshold of Table 3, and the ROPE (Region Of Practical Equivalence) probability. Where the analysis defined competing formulations, leave-one-out cross-validation (LOO) selects the one that better predicts unseen data: each observation is virtually removed in turn, the model is refitted on the remainder, and its prediction of the held-out point is scored. The formulation with the higher total score is preferred. The comparisons are:

- Model A across configuration-level covariate sets, testing whether overlap or patch size mitigates the effect (M1);
- Model B with and without probe-stage and geometry corrections (H1 representational);
- Model C with and without per-phase offsets, testing whether phase affects the deficit (H2);
- Model D1 across four grid labellings — stride only, patch only, both, neither — testing which branch generates the collapse sites (H3a, H3b).

Part 5 (Checks and verdicts). The final part tests whether each posterior can be trusted before interpreting it. The checks are not uniform across models because they target different risks:

- **Convergence** (all models): three preregistered diagnostics are verified for every fit. $\hat{R} < 1.01$ measures agreement across the four independent chains; values near 1 indicate that each chain has converged to the same distribution. Bulk and tail effective sample size above 1000 counts the genuinely independent posterior draws after correcting for autocorrelation. Zero divergences requires that the sampler encountered no geometric pathology. A model that fails any criterion is not interpreted.
- **Posterior predictive** (Model A): data are generated from the fitted posterior and compared against the observed contrasts, both pooled and stratified by configuration and generator. Model A is selected because it is the most complex and hierarchical; a simpler model that passes convergence is unlikely to fail this check.
- **Prior sensitivity** (Model A): the model is refitted across the scale ladder $\{0.25, 0.5, 1.0\}$ and with a Gaussian likelihood replacing the Student-t_4. The H1 conclusion is reportable only if the decision probability and credible interval remain stable across all variants; otherwise the conclusion reflects the choice of prior, not the data.
- **LOO reading rule**: the comparisons defined in Part 4 are collected. The key quantity is the difference in expected log pointwise predictive density (elpd_diff), which measures how much better one formulation predicts unseen data than the other; a difference smaller than two standard errors ($|\text{elpd\_diff}| < 2 \cdot \text{dse}$) is reported as inconclusive rather than rounded into a decision.

The notebook finally writes a verdict table applying the three-way rule of Table 3 to every hypothesis: supported, refuted or inconclusive.

Signal Generation and Testing

The experimental evaluation uses three synthetic signal generators: a single-harmonic sinusoidal generator, a modified TSMixup generator, and a KernelSynth generator. The last two are based on methods proposed for Chronos training.[6]

Synthetic Probing Setup

Single-Harmonic Sinusoidal Generator This generator produces a simple sinusoidal signal with a single frequency component. The generated signal can be expressed mathematically as:

$$x(t) = A \cdot \sin(2\pi f t + \phi) \tag{30}$$

The continuous signal is then sampled at a specified rate to create a discrete time-series. The parameters of the generator, such as amplitude, frequency, and phase, can be adjusted to create different sinusoidal signals for testing purposes.

Light TSMixup Generator We propose a controlled probing variant of the TSMixup method described in the Chronos data augmentation procedure.[6] The original TSMixup algorithm samples subsequences from real training datasets, applies mean scaling to each sampled time series, draws mixing coefficients from a symmetric Dirichlet distribution, and returns their convex combination.

Light TSMixup preserves this stochastic mixing structure, but replaces the real dataset collection with a controlled pool of sinusoidal signals. Each base signal is defined by frequency, amplitude, phase, length, and sampling frequency. This simplification makes the generated signals easier to interpret when testing whether Chronos-Bolt exhibits frequency-dependent blind spots or phase-locking effects.

The parameters of the generator, such as the maximum number of sinusoidal components, Dirichlet concentration, target length, and sampling frequency, can be adjusted to create different synthetic signals for testing purposes. The pseudocode for the Light TSMixup generator is listed in Appendix B.

KernelSynth Generator Since TSMixup is able to produce only simple composition of sinusoidal signals, we implement KernelSynth as a more complex alternative, which can generate signals with richer frequency content and temporal dynamics. By comparing the performance of Chronos-Bolt on signals generated by both methods, we can gain insights into how structural aliasing affects the model's ability to learn and forecast time-series data with varying levels of complexity.

Run	**Patch (P)**	**Stride (S)**	**overlap**	**tokens**
1	8	4	0.50	511
2	8	8	0.00	256
3	16	4	0.75	509
4	16	8	0.50	255
5	16	12	0.25	170
6	16	16	0.00	128
7	24	8	0.67	254
8	24	12	0.50	169
9	24	16	0.33	127
10	24	20	0.17	102
11	24	24	0.00	85
12	32	8	0.75	253
13	32	12	0.62	169
14	32	16	0.50	127
15	32	20	0.38	101
16	32	24	0.25	85
17	32	32	0.00	64

Table 4: *The extended configurations retained for the analysis. All are retrained from scratch under the same* 100*k-step budget and from the same seed (*42*).*

The KernelSynth generator is also described in the Chronos data augmentation procedure.[6] It produces synthetic signals by combining different kernel functions, using randomly selected weights from a normal distribution. The selected kernels are combined using sum and product operators. Its pseudocode, kernel bank and parameters are listed in Appendix C.

Experimental Protocol

All signals are sampled at $f_s = 512\,\text{Hz}$. Pure sinusoids are swept from 2 to 250 Hz in one-hertz steps at several non-redundant starting phases, so that every tone is fully representable under the Nyquist–Shannon theorem and a loss observed at a predicted lock frequency cannot be attributed to undersampling. A single seed (42) controls all stochastic generation, ensuring full reproducibility.

The complete experimental grid is built before any data are collected: one row per combination of geometry (P, S), lock frequency, background type, and phase. For each experimental condition, 200 background draws from both the Light TSMixup and KernelSynth generators are archived alongside the probing results, so that the exact inputs behind every posterior can be reloaded for verification. Each geometry is probed independently: the retrained Chronos-Bolt weights are loaded, the synthetic signals are generated, predictions are run, and the outputs are written to the observation tables that feed the Bayesian models of Bayesian Hypothesis.

The geometries listed in Table 4 are chosen to allow independent variation of P and S. Series at fixed P and varying S (P=16 with $S \in \{4, 8, 12, 16\}$; P=24 with $S \in \{8, 12, 16, 20, 24\}$; P=32 with $S \in \{8, 12, 16, 20, 24, 32\}$) isolate the stride effect, while series at fixed S and varying P (S=8 with $P \in \{8, 16, 24, 32\}$; S=12 with $P \in \{16, 24, 32\}$; S=16 with $P \in \{16, 24, 32\}$) isolate the patch effect. This factorial structure is what lets Models A and D of Bayesian Hypothesis separate the two branches and estimate the overlap and patch-size covariates without confounding.

Behavioral and Representational Analysis

The first analysis sweeps pure sinusoids from 2 to 250 Hz in one-hertz steps, at a sampling frequency of $f_s = 512\,\text{Hz}$. For each tone, the frequency produced by Chronos is compared with the known input frequency. Pure sinusoids keep the test easy to interpret: no background spectral content can explain a loss at a predicted lock frequency. The band stops below the Nyquist limit $f_s/2 = 256\,\text{Hz}$, so every tone swept is fully representable under the Nyquist–Shannon theorem, as Patch-Based Tokenization: a Formal Description sets out, and a loss observed here cannot be attributed to undersampling.

Table 5 summarises the patch size P, stride S, overlap $O = (P - S)/P$, and token count of each configuration. All models, including the P=S=16 baseline, are trained from scratch with the same 100k-step budget and seed; the published `chronos-bolt-tiny` checkpoint is not used. This keeps training length fixed across configurations while the token count shows the computational cost of overlap.

P	S	O	**tokens**	$S \mid P$	$P \mid S$
8	8	0.000	256	yes	yes
16	8	0.500	255	yes	**no**
16	12	0.250	170	**no**	**no**
16	16	0.000	128	yes	yes
24	24	0.000	85	yes	yes

Table 5: *The five initial geometries. The token count is the number of patch tokens at the 2048-sample training context. The last two columns are divisibility conditions: $S \mid P$ means every stride lock is also a patch null, and $P \mid S$ the converse. Both hold only at $P = S$, where the two branches are the same set of frequencies.*

Model	f_s/S [Hz]	f_s/P [Hz]	**stride only**	**both**	**patch only**
`p8-s8`	64	64	—	64 128 192	—
`p16-s8`	64	32	—	64 128 192	32 96 160 224
`p16-s12`	42.67	32	42.67 85.33 170.67 213.33	128	32 64 96 160 192 224
`p16-s16`	32	32	—	32 64 96 128 160 192 224	—
`p24-s24`	21.33	21.33	—	21.33 42.67 64 85.33 106.67 128 149.33 170.67 192 213.33 234.67	—

Table 6: *The locked frequencies of the five geometries of Table 5, with each frequency placed under the branch that produces it. A geometry with empty outer columns is one where $S \mid P$: the two branches coincide everywhere and no frequency is attributable to either alone.*

A locked frequency can satisfy the stride condition, the patch condition, or both. The notation $S \mid P$ means that P is an integer multiple of S. In that case every stride-lock site is also a patch-null site, so there are no stride-only sites, although patch-only sites may remain. Conversely, $P \mid S$ means that every patch-null site is also a stride-lock site, so there are no patch-only sites. If neither relation holds, both types of exclusive site are possible; if $P = S$, the two sets coincide.

Table 6 applies this rule to the sweep by sorting each candidate frequency into stride only, both, or patch only. Its main purpose is attribution: a dip at an exclusive site can be associated with one branch, whereas a dip at a shared site cannot separate their contributions.

Each configuration is swept over the same band at several non-redundant starting phases. The nominal 512-sample context is shortened, when necessary, to a whole number of strides so that the final patch is not padded; otherwise padding could create an artificial collapse. When the stride divides 512, integer-frequency tones also complete whole cycles in the window. For other strides, the context differs by fewer than S samples. Each model is loaded, measured, and released in turn.

Frequency Reconstruction

This reading is behavioural and records what the model outputs. The sinusoid is given as context, the median forecast is appended and fed back, and the rollout continues to 512 generated samples. Its dominant frequency is read as the largest magnitude of the discrete Fourier transform after the mean is removed and the zero-frequency bin skipped, then averaged over phases and plotted against the input frequency. A point on the diagonal is a tone rebuilt correctly, a flat segment is a set of distinct inputs collapsing onto one output, and the shaded band is one standard deviation over phases.

Figure 1 plots the dominant frequency of the rollout against the frequency of the input tone. The dotted diagonal is a perfect reconstruction and the dashed verticals are $\mathcal{F}_{lock}$.

The P=16, S=16 geometry is that of the published Chronos-Bolt checkpoint, but only the setting is shared: the model evaluated here is not that checkpoint; it is a variant retrained from scratch under the same 100k-step budget. Its reconstruction curve leaves the diagonal over wide stretches of the band. The figure does not licence a claim about structural aliasing: it is behavioural, and records what the model outputs rather than what its representation retains.

Figure 2 repeats the reading on the other four geometries, and the pattern is the same in each of them: every model rebuilds some frequencies correctly while others are not rebuilt at all.

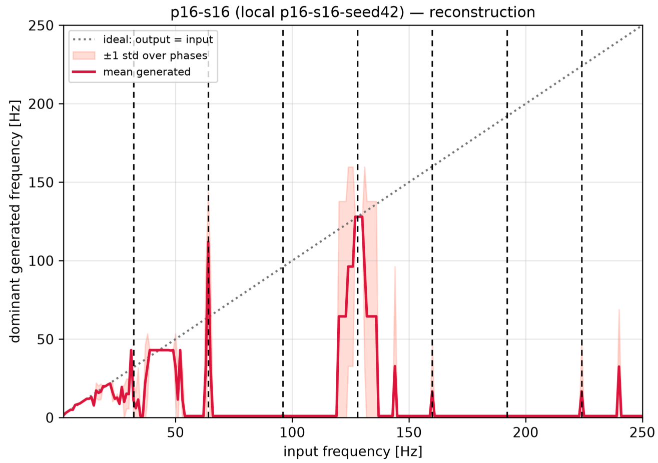

Figure 1: *Reconstruction on the $P = 16, S = 16$ geometry: dominant frequency of the rollout against the tone supplied, averaged over phases, one standard deviation shaded. The dotted diagonal is a tone rebuilt correctly; the dashed verticals are $\mathcal{F}_{lock}$.*

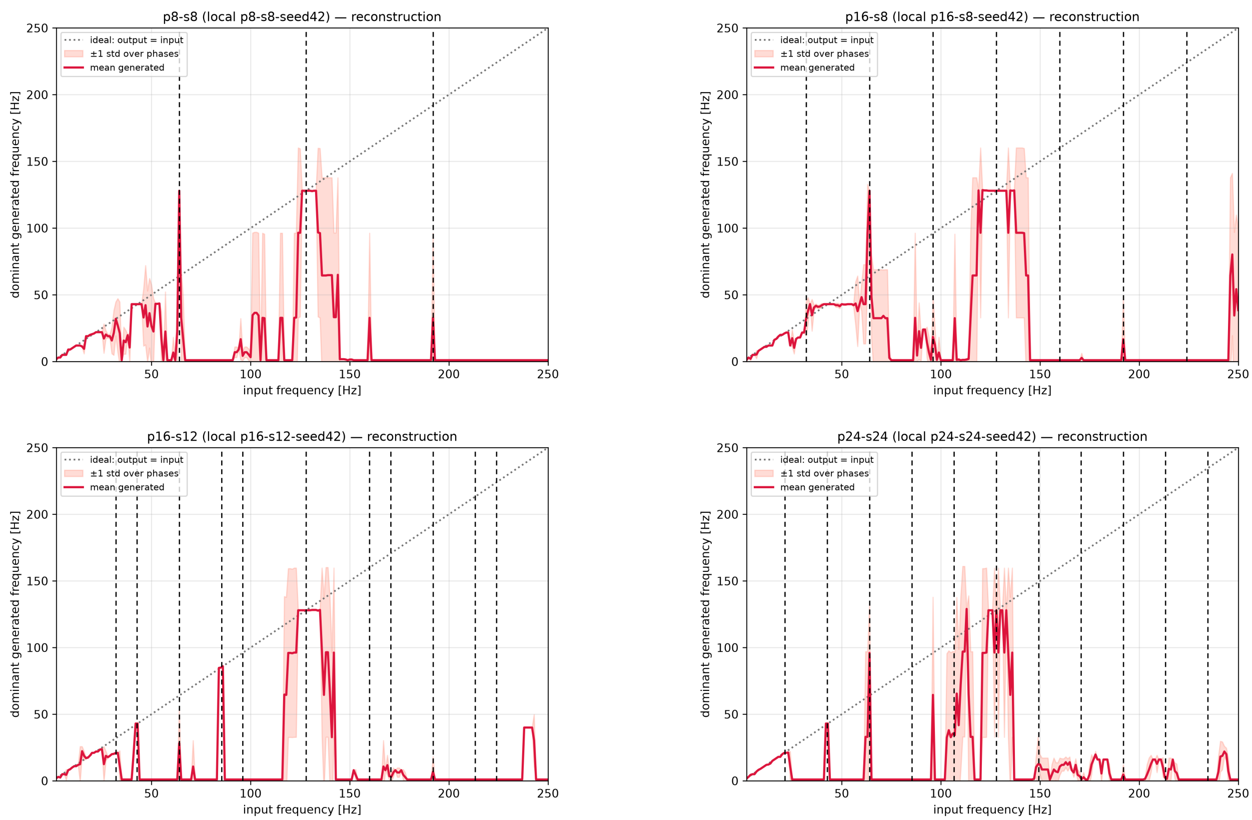

Figure 2: *Reconstruction on the other four geometries: dominant frequency of the rollout against the tone supplied, averaged over phases, one standard deviation shaded. The dotted diagonal is a tone rebuilt correctly; the dashed verticals are $\mathcal{F}_{lock}$.*

Spectral Recovery

This reading is representational and records what the model still represents. The internal state is captured at the projected quantile head, and a linear probe (standardisation, then a principal component reduction, then logistic regression) is scored by cross-validation. The reading here is taken at that one stage; Model B of Bayesian Hypothesis repeats it at ten. Two task families are run. The seven band tasks each ask whether a tone falls in the lower or upper half of its band; their folds are grouped by frequency, so every phase of a held-out frequency is excluded from training and the probe has to generalise the boundary instead of memorising individual frequencies. An ungrouped split inflates those rows. The local task poses one question of constant difficulty at every frequency, whether $f - 1\,\mathrm{Hz}$ can be separated from $f + 1\,\mathrm{Hz}$, and holds out phases rather than frequencies; having no boundary artefact, it is the row the cross-geometry comparison uses. Cell colour is that cross-validated accuracy on a continuous scale from chance to one, so a partial loss is rendered as such rather than rounded into a bin; grey marks a frequency no task covers, which is a statement about the design and not about the model. Dashed markers on both panels are $\mathcal{F}_{lock}$ for that geometry.

Figure 3 and Figure 4 ask the same band of the representation instead of the forecast. In the local task the accuracy drops are narrow and fall close to some of the predicted lock frequencies; in the band tasks the drops instead extend over a range of frequencies around them.

These observations do not establish structural aliasing by themselves, but they do show that the models represent certain frequencies with more difficulty than others. They are the empirical ground on which the Bayesian models of Bayesian Hypothesis are built. The Bayesian fits share a fixed 480-sample probing context across geometries, so a stride that does not divide it can be swept but not fitted.

Limits of this design

The five geometries of Table 5 bound what can be concluded from the sweep above. Three of them have $P = S$, so their two branches coincide at every locked frequency and no observation in those geometries can be attributed to one branch. Stride-only sites exist in one geometry, P=16, S=12, and patch-only sites in two, so a claim about the stride branch rests on a single configuration and a claim about the patch branch on two. The configuration axes are equally short: the overlap ratio takes three values, and exactly one pair shares a stride while differing in patch size, which is the only place an effect of the patch can be separated from an effect of the stride. Every statement is therefore conditional on a narrow design. Attributing an effect to one branch, and testing whether either branch moves with the spacing it predicts, is beyond what these five runs can support: they can show that some frequencies are lost, not which of the two grids accounts for them. Both properties require the extended set of Table 4.

Bayesian Results

The frequency sweeps and the models of Bayesian Hypothesis point in the same direction: phase-locking tends to affect the internal representation but does not appear to propagate to the forecast output.

On the behavioural side, locked frequencies are not systematically attenuated in the rollout. In several geometries the model reconstructs them as well as their unlocked neighbours, contrary to what the degeneracy alone would predict. The representational side points toward a different pattern: the prequential codelength at a locked frequency tends to be higher than at an unlocked one, suggesting that the internal state carries a measurable cost at those sites. This split, representation penalised and forecast intact, is the pattern the evidence points to, and the per-generator and per-geometry evidence follows.

The phase of the input signal does not appear to modulate the deficit appreciably, consistent with $H2$: the posterior for σ_ϕ in Figure 5 lies inside the region of practical equivalence (ROPE). It concentrates well below the threshold and is far tighter than the prior it started from, so the data and not the prior place it there, and the eight per-phase offsets u_p are not separable from one another. The size of the deficit therefore follows from the geometry and from the frequency, and not from where in its cycle the signal begins, which is

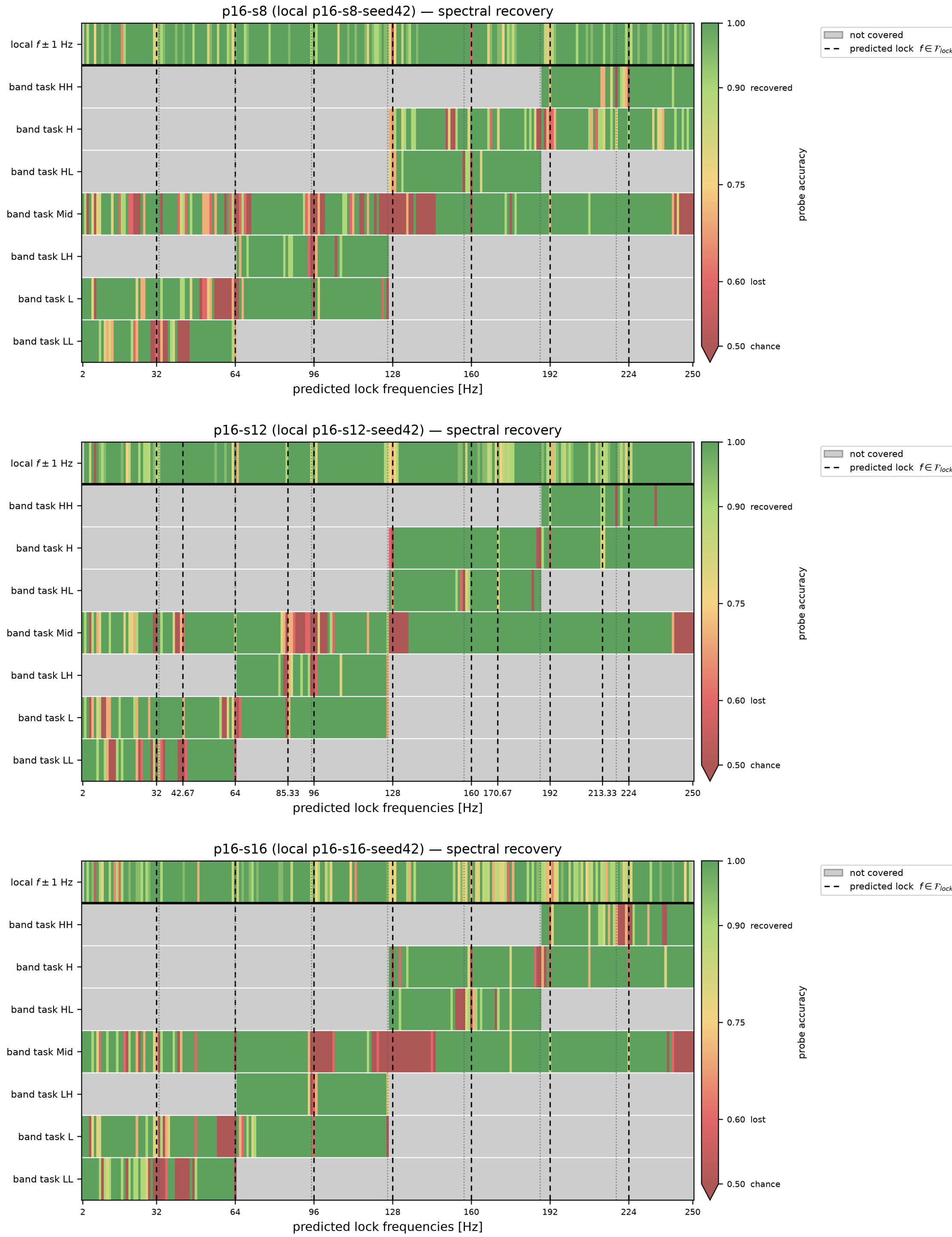

Figure 3: *Spectral recovery at $P = 16$, the stride lengthening from $S = 8$ to $S = 16$, so a feature that moves between panels belongs to the stride branch. Rows are the seven band tasks and, above the rule, the local $f \pm 1$ Hz task; colour is probe accuracy from chance to one, grey a frequency the task does not cover, dashed verticals $\mathcal{F}_{lock}$.*

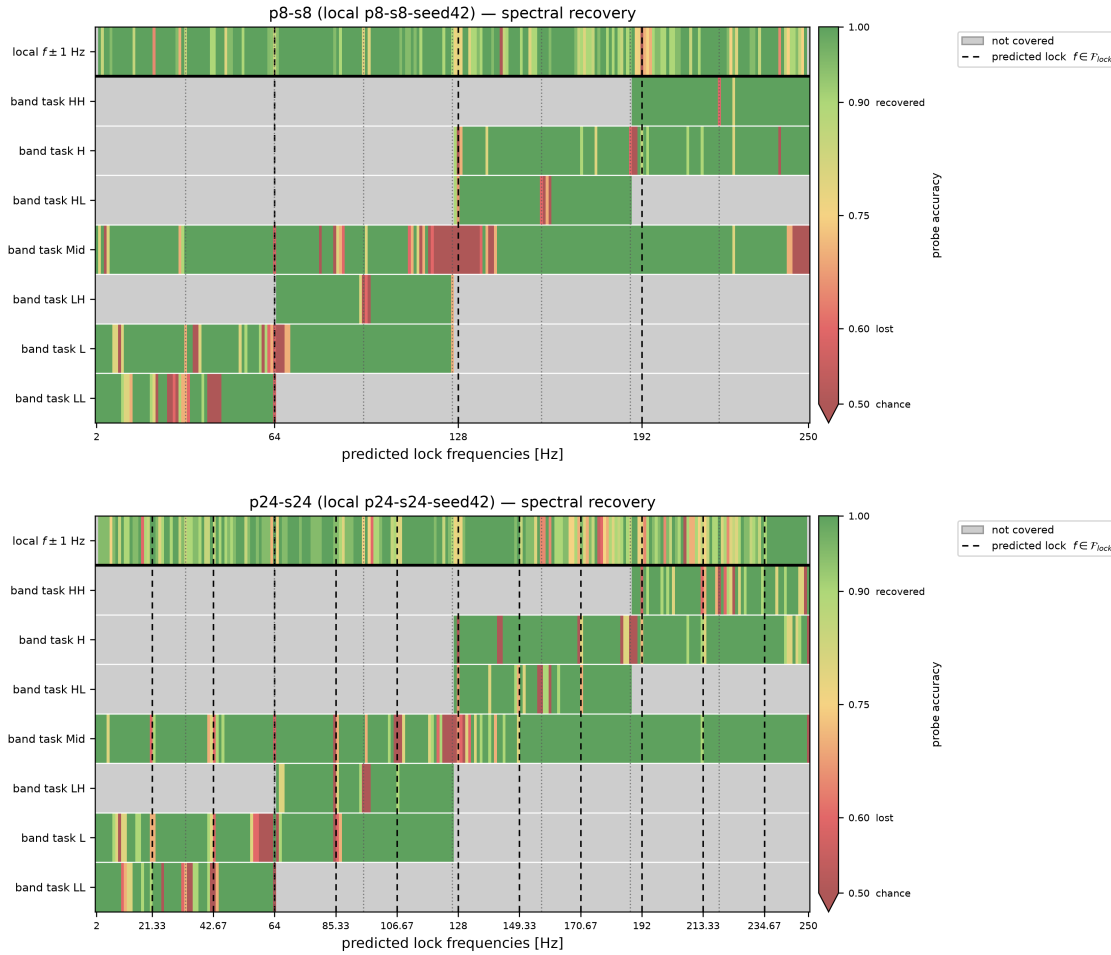

Figure 4: *Spectral recovery at the two $P = S$ geometries, where the patch length varies at zero overlap and the two branches coincide in both. Rows are the seven band tasks and, above the rule, the local $f \pm 1$ Hz task; colour is probe accuracy from chance to one, grey a frequency the task does not cover, dashed verticals $\mathcal{F}_{lock}$.*

what licences the remaining models to average over phase and which also removes phase randomisation from the mitigations available to an operator.

The token-collapse dips align with the stride grid and shift when the stride changes, consistent with $H3a$, and Figure 6 shows the pattern holding across configurations and generators; the scaling law of the patch branch is not identified in the runs fitted for this deliverable.

Models A and C have not reached the preregistered convergence thresholds; their estimands are nonetheless consistent with the converged representational model (B) and with the independent frequentist sweeps, so the qualitative conclusions hold provisionally. Consolidated posteriors or, if convergence remains out of reach, a documented limitation and fall-back on the frequentist evidence will follow.

Mitigation Strategy

The mitigation carried through the design is increased patch overlap: shorter S at fixed P. Its mechanism is arithmetic: the stride branch has its lowest in-band member at f_s/S, so halving the stride doubles that frequency and thins the family. The same arithmetic bounds what it can achieve. Overlap does not affect the patch branch at all, so it thins at most one of the two. On the one-minute telemetry of the photovoltaic mapping the k-th member of that branch is a component of period P/k minutes, and those periods stay blind. At a fixed context the token count also grows as $1/S$, which is a real constraint for an edge-deployed forecaster.

Both halves are testable rather than asserted, and $M1$ in Table 3 is the test. Whether alternative mitigations can improve on this bound is discussed below.

Agentic AI

Computer Security

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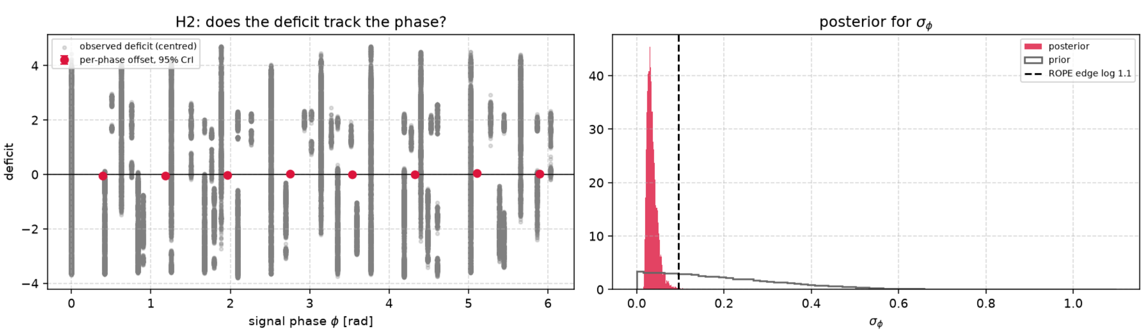

Figure 5: *Posterior for the phase term of H2. On the left the observed deficit against the signal phase ϕ, with the per-phase offset u_p and its 95% credible interval; on the right the posterior for σ_ϕ against its prior, the dashed line marking the ROPE edge at $\log 1.1$. Both panels pool every configuration fitted, so the estimate is not conditioned on a single geometry. Model C has not met the convergence thresholds of Bayesian Hypothesis, so the posterior is shown as provisional.*

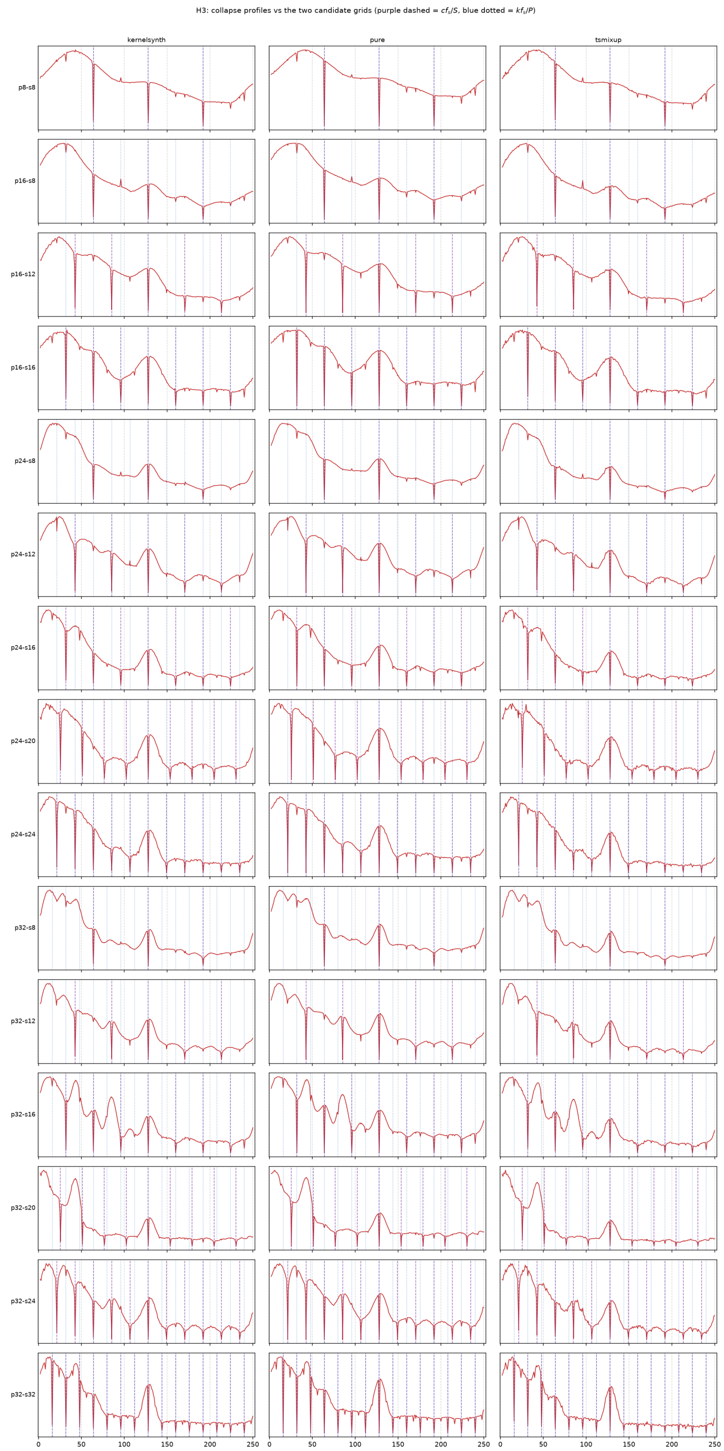

Figure 6: *Collapse profiles against the two candidate grids. Rows are the fitted configurations and columns the three generators; each panel plots the profile over the sweep band, with the stride grid cf_s/S dashed and the patch grid kf_s/P dotted. A dip that sits on the dashed grid and follows it as the stride changes belongs to the stride branch. The run covers every configuration of Table 4.*

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Appendix A: Ontology Representation

The core ontology architecture bridges the gap between cyber-physical energy systems and deep learning signal processing domain knowledge. It categorizes the application setting into five main conceptual pillars rooted under the ontological hierarchy:

- **Physical Subsystem (`pv:PhysicalComponent`):** Represents the hardware infrastructure of the micro-grid, including the solar array, electrochemical storage, consumer loads, and the macro grid tie.
- **Cognitive Layer (`pv:ControlAgent`):** The artificial intelligence system responsible for autonomous grid optimization, load balancing, dispatch scheduling, and proactive vulnerability mitigation.
- **Signal Processing Layer (`pv:Signal`, `pv:PatchedSignal`):** Tracks the raw telemetry inputs (both time-domain features and frequency-domain components) along with their tokenized representation configurations (patch dimensions, stride values) tailored for the Chronos foundation architecture.
- **Environmental Tracking (`pv:WeatherObservation`):** Captures concurrent meteorological snapshots (air temperature, wind vectors) influencing structural physics and electrical efficiency.
- **Inferred Operational States:** Dynamic metadata classified into generation levels (`pv:GenerationState`), systemic disruptions (`pv:AnomalyState`), control directives (`pv:AgentAction`), and signal degradation hazards (`pv:AliasingEvent`).

Core Concept Taxonomy

The formal classes defined within the schema are explicitly detailed below:

`pv:PhotovoltaicArray` The primary renewable energy asset. Characterized by panel count and nameplate nominal capacity. Source of telemetry streams; associated with weather conditions and subject to environmental/electrical anomalies.

`pv:BatteryStorage` Local electrochemical reservoir capable of dynamic grid connection/isolation. Characterized by capacity and instantaneous State of Charge (SoC).

`pv:InternalLoad` Represents internal industrial or residential consumption demands. Subject to adjustable, demand-side management interventions.

`pv:NationalGrid` The external electrical macro-grid interface, acting as a bidirectional infinite bus for excess export or deficit imports.

`pv:Signal` Time-series measurement stream containing global horizontal irradiance (G_h), tilted surface irradiance (G_{tilt}), instantaneous power output (P_{pv}), and analytical frequency descriptors (sampling rate, fundamental frequency, amplitude, phase).

`pv:PatchedSignal` The Chronos-specific tokenized structure derived from a raw signal. Governed by a selected window chunking mechanism (*patchSize*, *stride*), establishing a structural *overlapRatio* and a downstream $patchInducedFrequency$.

Semantic Relations & Properties

Object Properties (Entity-to-Entity Relationships)

The physical, functional, and inferred linkages across concepts are formalized using the following object properties:

- ⟨`pv:PhotovoltaicArray`⟩ `pv:measures` ⟨`pv:Signal`⟩
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasWeatherObservation` ⟨`pv:WeatherObservation`⟩
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasGenerationState` ⟨`pv:GenerationState`⟩ *(Inferred via SWRL)*
- ⟨`pv:PhotovoltaicArray`⟩ `pv:hasAnomaly` ⟨`pv:AnomalyState`⟩ *(Inferred via SWRL)*
- ⟨`pv:PhotovoltaicArray`⟩ `pv:triggersAction` ⟨`pv:AgentAction`⟩ *(Inferred via SWRL)*

- $\langle$`pv:Signal`$\rangle$ `pv:tokenizes` $\langle$`pv:PatchedSignal`$\rangle$
- $\langle$`pv:PatchedSignal`$\rangle$ `pv:exhibitsAliasing` $\langle$`pv:AliasingEvent`$\rangle$ *(Inferred via SWRL)*
- $\langle$`pv:BatteryStorage`$\rangle$ `pv:buffers` $\langle$`pv:PhotovoltaicArray`$\rangle$
- $\langle$`pv:PhysicalComponent`$\rangle$ `pv:connectedTo` $\langle$`pv:PhysicalComponent`$\rangle$ *(Symmetric Property)*

Data Properties (Entity Attributes)

Key quantitative attributes defining system conditions include:

- **Electrical/Solar:** `pv:pvPowerW` (P_{pv}), `pv:nominalPowerW` (P_{nom}), `pv:tiltedIrradianceWm2` (G_{tilt}), `pv:deltaPowerW` (ΔP).
- **Meteorological:** `pv:airTemperatureC` (T_{air}), `pv:windSpeedMs` (W_s).
- **Tokenization Architecture:** `pv:patchSize` (N_p), `pv:stride` (S), `pv:overlapRatio` (γ), `pv:patchInducedFrequency` (f_p).
- **Signal Descriptors:** `pv:samplingFrequencyHz` (f_s), `pv:fundamentalFrequencyHz` (f_0), `pv:phaseRad` (ϕ).

Knowledge-Driven Inference System (SWRL Rules)

Real-time diagnostic, classification, and mitigation logic are automated using a rule-based engine. The standard logical form of these production rules is detailed below:

Environmental Context Extraction

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \geq 800.0) \\ &\longrightarrow \texttt{hasGenerationState}(?pv, \texttt{ClearSky})\end{aligned} \tag{31}$$

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \\ &\wedge (?g \geq 150.0) \wedge (?g < 800.0) \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{PartialCloud})\end{aligned} \tag{32}$$

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \\ &\wedge (?g > 0.0) \wedge (?g < 150.0) \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Overcast})\end{aligned} \tag{33}$$

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g \leq 0.0) \\ &\longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Nighttime})\end{aligned} \tag{34}$$

Cyber-Physical Anomaly Diagnostics

- **Inverter Clipping Anomaly (R5):**

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{pvPowerW}(?s, ?p) \wedge \texttt{nominalPowerW}(?pv, ?nom) \\ &\wedge (?p \geq ?nom) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 800.0) \\ &\longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Clipping})\end{aligned} \tag{35}$$

- **Inverter Trip Anomaly (R6):**

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{pvPowerW}(?s, ?p) \wedge (?p \leq 0.0) \\ &\wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 200.0) \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{InverterTrip})\end{aligned} \tag{36}$$

- **Ramp Event Anomaly (R7):**

$$\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{deltaPowerW}(?s, ?dp) \wedge (?dp > 500.0) \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Ramp}) \tag{37}$$

- **Soiling Performance Deficit Anomaly (R8):**

$$\begin{aligned} &\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 400.0) \\ &\quad \wedge \texttt{pvPowerW}(?s, ?p) \wedge (?p > 100.0) \wedge \texttt{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.3) \\ &\quad \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Soiling}) \end{aligned} \tag{38}$$

- **Thermal Overheating Derating Anomaly (R9):**

$$\begin{aligned} &\texttt{PhotovoltaicArray}(?pv) \wedge \texttt{hasWeatherObservation}(?pv, ?w) \wedge \texttt{airTemperatureC}(?w, ?t) \wedge (?t > 35.0) \\ &\quad \wedge \texttt{measures}(?pv, ?s) \wedge \texttt{tiltedIrradianceWm2}(?s, ?g) \wedge (?g > 600.0) \\ &\quad \wedge \texttt{pvPowerW}(?s, ?p) \wedge \texttt{swrlb:divide}(?r, ?p, ?g) \wedge (?r < 0.4) \\ &\quad \longrightarrow \texttt{hasAnomaly}(?pv, \texttt{Overheating}) \end{aligned} \tag{39}$$

Autonomous Closed-Loop Control Actuation

Based on inferred operating profiles, actions are dynamically mapped to handle grid stresses:

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:Ramp}) \longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:RampAlert}) \tag{40}$$

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:Clipping}) \vee \texttt{pv:hasAnomaly}(?pv, \texttt{pv:Overheating}) \tag{41}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:CurtailGeneration}) \tag{42}$$

$$\texttt{pv:hasAnomaly}(?pv, \texttt{pv:InverterTrip}) \vee \texttt{pv:hasGenerationState}(?pv, \texttt{pv:PartialCloud}) \tag{43}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:BatteryDispatch}) \tag{44}$$

$$\texttt{pv:hasGenerationState}(?pv, \texttt{pv:Overcast}) \vee \texttt{pv:hasGenerationState}(?pv, \texttt{pv:Nighttime}) \tag{45}$$

$$\longrightarrow \texttt{pv:triggersAction}(?pv, \texttt{pv:LoadAdjustment}) \tag{46}$$

Chronos Structural Aliasing Hypotheses

The key scientific contribution of this architecture lies in identifying information loss bounds during time-series patching. The model detects structural blind spots via frequency domain reasoning:

Hypothesis H1 & H1b: Patch-Induced Frequency Blind Spots

The patch tokenization process introduces an artificial operating frequency profile f_p, mathematically defined by the relationship:

$$f_p = \frac{f_s}{S} \tag{47}$$

Where f_s represents the raw Nyquist sampling rate and S denotes the token step stride.

The appropriate detection tolerance derives from the frequency resolution of the context window of length N_{context} samples:

$$\delta f \approx \frac{f_s}{N_{\text{context}}}. \tag{48}$$

For Chronos-Bolt default context lengths (512–2048 samples), this gives $\delta f \in [f_s/2048,\ f_s/512]$. The tolerance fraction $\varepsilon = \delta f / f_p = S/N_{\text{context}}$ must be validated empirically; whether $\pm 10\%$ is wider or narrower than this resolution depends on the specific S and N_{context} chosen.

- **Primary Blind Spot (H1):**

$$\begin{aligned}&\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{fundamentalFrequencyHz}(?s, ?f_0) \wedge \texttt{patchInducedFrequencyHz}(?ps, ?f_p) \\ &\qquad \wedge \texttt{swrlb:divide}(?r, ?f_0, ?f_p) \wedge (?r \geq 1 - \varepsilon) \wedge (?r \leq 1 + \varepsilon) \\ &\qquad \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H1\_BlindSpot})\end{aligned} \tag{49}$$

- **Secondary Harmonic Blind Spot (H1b):**

$$\begin{aligned}&\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{fundamentalFrequencyHz}(?s, ?f_0) \wedge \texttt{patchInducedFrequencyHz}(?ps, ?f_p) \\ &\qquad \wedge \texttt{swrlb:divide}(?r, ?f_0, ?f_p) \wedge (?r \geq 2 - \varepsilon) \wedge (?r \leq 2 + \varepsilon) \\ &\qquad \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H1b\_BlindSpot})\end{aligned} \tag{50}$$

Hypothesis H2: Phase Aliasing Effect

Vulnerabilities appear when temporal offsets misalign with chunk window anchors, corrupting early representation extraction features ($0 < \phi < S$):

$$\begin{aligned}&\texttt{Signal}(?s) \wedge \texttt{tokenizes}(?s, ?ps) \wedge \texttt{phaseRad}(?s, ?\phi) \wedge \texttt{stride}(?ps, ?S) \\ &\qquad \wedge (?\phi > 0.0) \wedge (?\phi < ?S) \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H2\_Phase})\end{aligned} \tag{51}$$

Hypothesis H3: Overlap Information Distortion

When the overlap configuration ratio ($\gamma = \frac{N_p - S}{N_p}$) drops below a critical information-theoretic bound ($\gamma < 0.5$), consecutive contextual patches display heavy text-token gaps, rendering trend forecasts highly unstable:

$$\texttt{PatchedSignal}(?ps) \wedge \texttt{overlapRatio}(?ps, ?or) \wedge (?or < 0.5) \longrightarrow \texttt{exhibitsAliasing}(?ps, \texttt{H3\_Overlap}) \tag{52}$$

Appendix B: Light TSMixup Pseudocode

Algorithm 1 Light TSMixup: Controlled Sinusoidal Time Series Mixup

Require: Sinusoidal pool $\mathcal{P} = \{p_1, \ldots, p_{N_p}\}$, where each p_n is defined by frequency f_n, amplitude A_n, phase ϕ_n, source length T_n, and sampling frequency f_s; maximum components $K = 10$; symmetric Dirichlet concentration $\alpha = 1.5$; minimum and maximum augmented lengths ($l_{\min} = 128, l_{\max} = 2048$).

Ensure: An augmented time series.

$k \sim U\{1, K\}$ ▷ number of sinusoidal components to mix

$l \sim U\{l_{\min}, l_{\max}\}$ ▷ length of the augmented time series

for $i \leftarrow 1$ to k **do**

  $n \sim U\{1, N_p\}$ ▷ sample a sinusoidal source

  $x_{1:l}^{(i)} \leftarrow \mathrm{SampleSegment}(p_n, l, f_s)$ ▷ generate or crop a length-l segment

  $\tilde{x}_{1:l}^{(i)} \leftarrow x_{1:l}^{(i)} \Big/ \left(\frac{1}{l} \sum_{j=1}^{l} \left| x_j^{(i)} \right| \right)$ ▷ apply mean scaling

end for

$[\lambda_1, \ldots, \lambda_k] \sim \mathrm{Dir}([\alpha_1 = \alpha, \ldots, \alpha_k = \alpha])$ ▷ sample mixing weights

return $\sum_{i=1}^{k} \lambda_i \tilde{x}_{1:l}^{(i)}$ ▷ take a weighted combination of scaled signals

Appendix C: KernelSynth Pseudocode and Kernel Bank

Algorithm 2 KernelSynth: Synthetic Data Generation using Gaussian Processes

Require: Kernel bank $\mathcal{K}$ listed in Table 7; maximum kernels per time series $J = 5$; generated length $l_{\text{syn}} = 1024$; optional numerical jitter $\epsilon = 0$.

Ensure: A synthetic time series $x_{1:l_{\text{syn}}}$.

$j \sim U\{1, J\}$ ▷ sample the number of kernels

$\{\kappa_1(t,t'), \ldots, \kappa_j(t,t')\} \overset{\text{i.i.d.}}{\sim} \mathcal{K}$ ▷ sample kernels with replacement

$t \leftarrow \text{linspace}(0, 1, l_{\text{syn}})$

$\kappa^\star(t,t') \leftarrow \kappa_1(t,t')$ ▷ initialize the composite covariance

for $i \leftarrow 2$ to j **do**

  $\star \sim U\{+, \times\}$ ▷ sample a binary composition operator

  $\kappa^\star(t,t') \leftarrow \kappa^\star(t,t') \star \kappa_i(t,t')$ ▷ compose kernels

end for

if $\epsilon > 0$ **then**

  $\kappa^\star(t,t') \leftarrow \kappa^\star(t,t') + \epsilon I$

end if

$x_{1:l_{\text{syn}}} \sim \mathcal{GP}(0, \kappa^\star(t,t'))$ ▷ sample from the GP prior

return $x_{1:l_{\text{syn}}}$

Table 7: *Kernel bank $\mathcal{K}$ used by KernelSynth.*

Kernel	Formula	Hyperparameters
Constant	$\kappa_{\text{Const}}(x,x') = C$	$C = 1$
White Noise	$\kappa_{\text{White}}(x,x') = \sigma_n \cdot \mathbf{1}_{x=x'}$	$\sigma_n \in \{0.1, 1\}$
Linear	$\kappa_{\text{Lin}}(x,x') = \sigma_0^2 + x \cdot x'$	$\sigma_0 \in \{0, 1, 10\}$
RBF	$\kappa_{\text{RBF}}(x,x') = \exp\left(-\frac{\lVert x-x'\rVert^2}{2l^2}\right)$	$l \in \{0.1, 1, 10\}$
Rational Quadratic	$\kappa_{\text{RQ}}(x,x') = \left(1 + \frac{\lVert x-x'\rVert^2}{2\alpha}\right)^{-\alpha}$	$\alpha \in \{0.1, 1, 10\}$, $l = 1$
Periodic	$\kappa_{\text{Per}}(x,x') = \exp\left(-2\sin^2\left(\pi\frac{\lVert x-x'\rVert}{p/l_{\text{syn}}}\right)\right)$	$p \in \{24, 48, 96, 168, 336, 672,$ $7, 14, 30, 60, 365, 730,$ $4, 26, 52, 4, 6, 12, 4, 40, 10\}$